

Performance-Focused Design Exercise Packet

Part of ASME IDETC 2025 Workshop:

Adventures in Tradespace Exploration

17 August 2025

Group ID: _____



Annotate, mark-up, and write on this page as needed.

Round 1:

For this design exercise, you will be acting as a member of your assigned working group, and you will be provided cross-disciplinary performance space plots.

INSTRUCTIONS:

Working in your assigned working group, select a single solution using the performance space plots provided in the supplemental packet. You may annotate and mark the supplemental packet as you wish. You wish to maximize the vehicle's performance in your own objectives, and the other working group wishes to maximize their own objectives, but you must select a single vehicle from the 20 designs given in the supplemental packet. Write the selected design's ID in the space provided in the supplemental packet.

The Suspension and Driveline working group wishes to maximize the following objectives:

- A. Acceleration**
- B. Off Road Ability**

10, 1, 20, 11

The Chassis and Cabin working group wishes to maximize the following objectives:

- C. Occupant Protection**
- D. Passenger Cabin Space**

8, 4, 11

1 1

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Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables.

107"	87.6"	78"	8.1"	750	2.931	30"	880	2.335"	1.97 _{in}
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Do you think you could have found a better solution given more time?

Yes.

What, in this round, stopped you from finding a better solution?

- time. With more time we could dive into each individual thing
- A customer prioritization list for work currently works well and what they would specifically want to change

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Round 2:

For this design exercise, you will be acting as a member of your assigned working group – you will have differing objectives and be provided performance space plots that DO NOT have any cross-disciplinary objective pairings.

INSTRUCTIONS:

Working in your assigned working group, select a single solution using the performance space plots provided in the supplemental packet. You may annotate and mark the supplemental packet as you wish. You wish to maximize the vehicle's performance in your own objectives, and the other working group wishes to maximize their own objectives, but you must select a single vehicle from the 20 designs given in the supplemental packet. Write the selected design's ID in the space provided in the supplemental packet.

The Suspension and Driveline working group wishes to maximize the following objectives:

E. Top Speed

F. Towing Capacity

20, 5, 9, 11

The Chassis and Cabin working group wishes to maximize the following objectives:

G. Low Speed Maneuverability

H. Rollover Stability

3, 13, 11, 20, 4

4

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Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables.

181"	81.2"	69.2"	9.7"	655	237.1	32"	1070	0.877	1.99'
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Do you think you could have found a better solution given more time?

No. We had a good foundation

What, in this round, stopped you from finding a better solution?

Not looking at the old graphs from round 1.
We don't have a top speed ~~to~~

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Round 3:

For this design exercise, you will NOT be acting as a member of your assigned working group. You will act as a team that shares the same objectives.

INSTRUCTIONS:

Working as a cross-disciplinary team, select a single solution using the performance space plots provided in the supplemental packet. You may annotate and mark the supplemental packet as you wish. Select a single vehicle from the 20 designs presented and write the selected design's ID in the space provided in the supplemental packet.

Your team wishes to maximize the following objectives:

- 7 6 132 - I. Off Road Ability - 2, 5, 10, 5, 10
6 13 7 - J. Operational Range - 2, 5, 10, 6, 3
6 13 7 - K. Maintainability - 3, 6, 13, 10, 6, 3
6 7 13 - L. Passenger Cabin Space - 18, 17, 15

7 balanced
6

6
12
24

6

6 8
12 4
6 1
3 3

27 16

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Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables

178"	811"	791"	6.8"	409	253.1	47"	1530	2.257	2.47
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Do you think you could have found a better solution given more time?

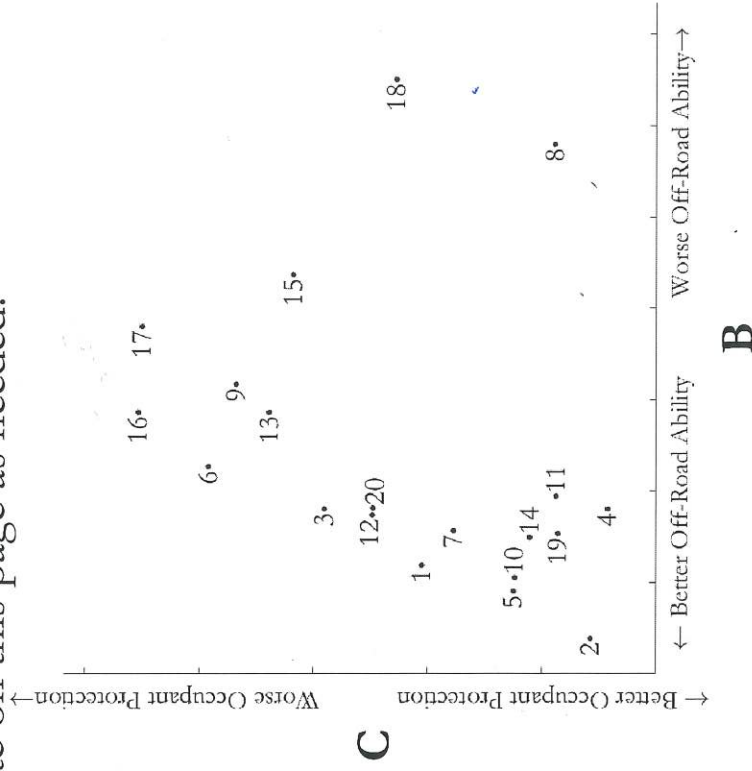
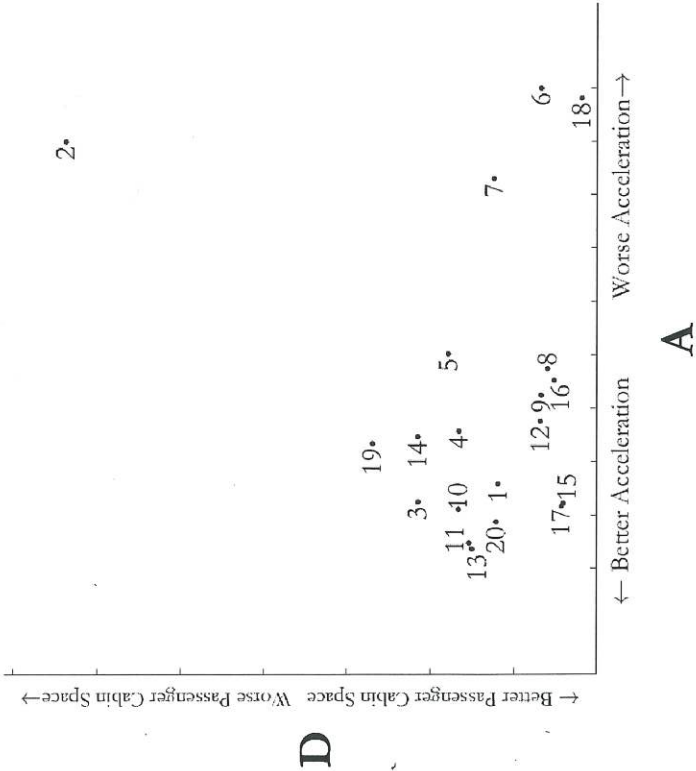
Yes

What, in this round, stopped you from finding a better solution?

Time to do 2FD Math

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Notes:
• We are prioritizing occupancy protection

Grouping Relations

	A	B	C	D
A				
B				
C				
D				

A-B: 10, 1, 20, 11
C-D: 8, 4, 11

Individual "Worst Options"

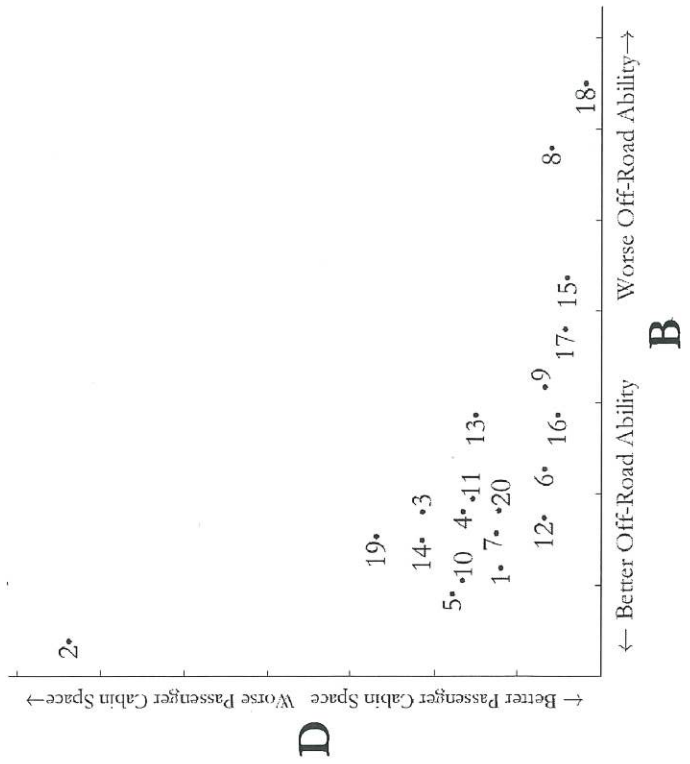
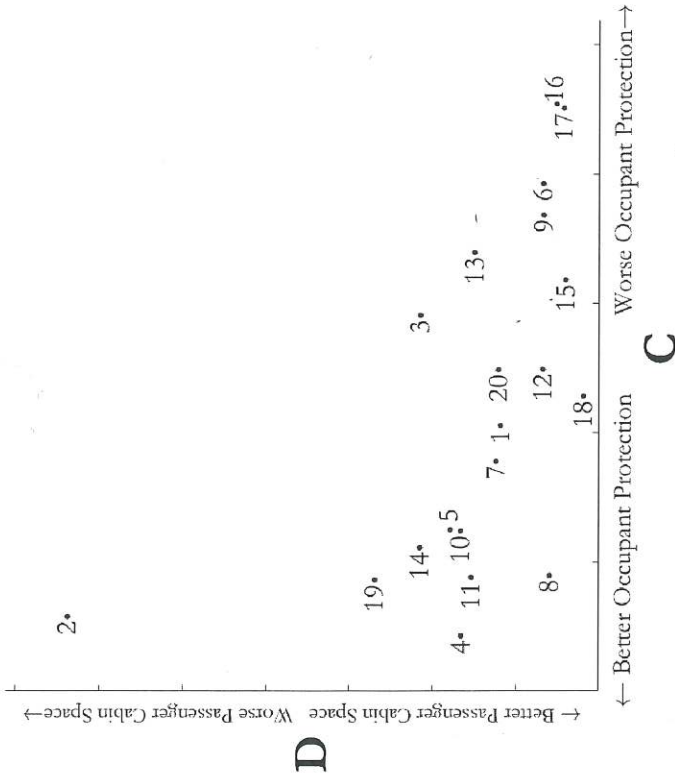
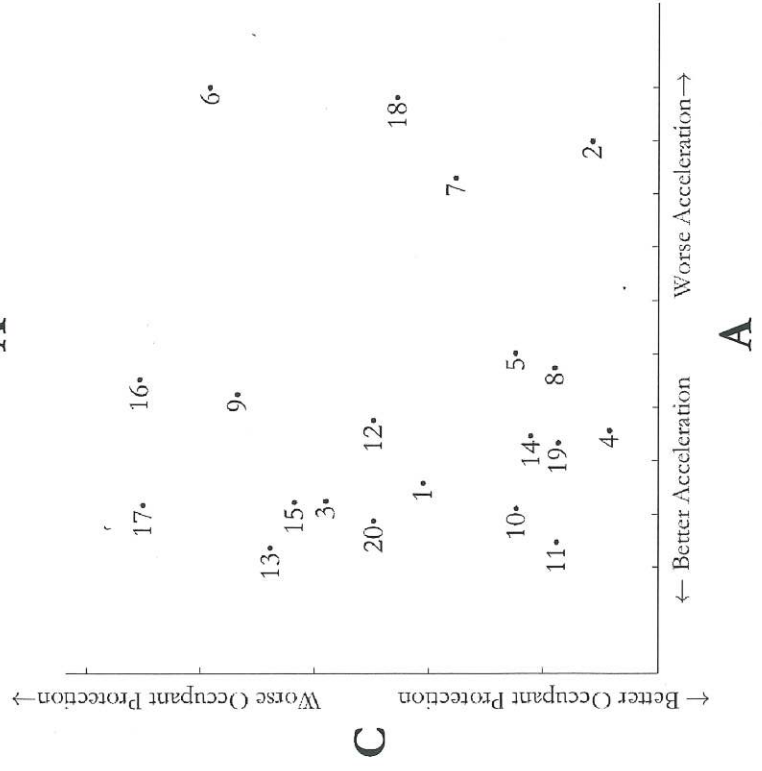
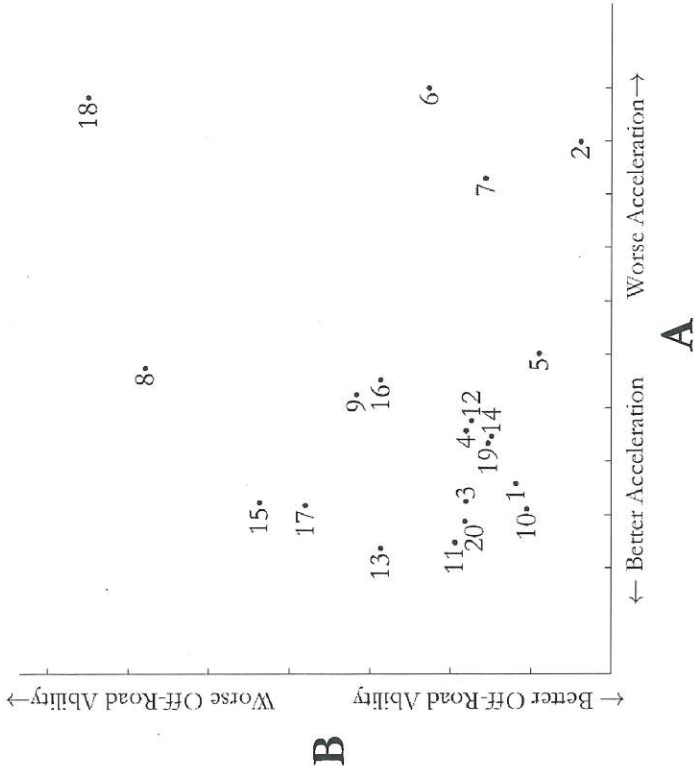
A: 13 B: 2 C: 4 D: 18

Individual "Best Options"

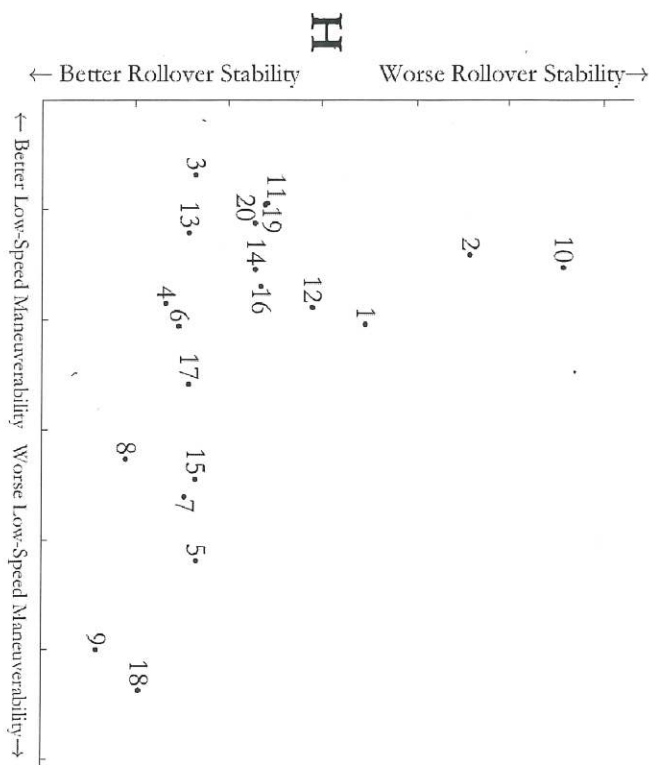
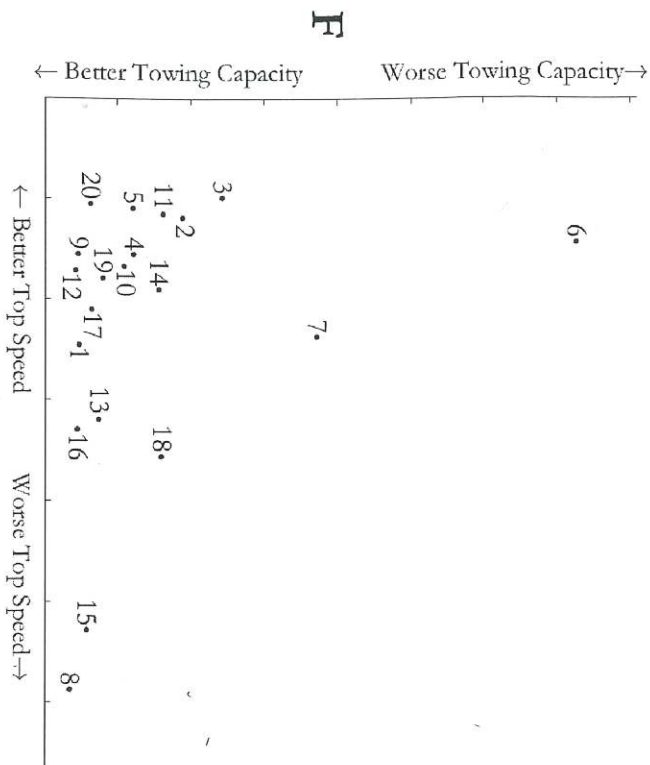
Enter the number ID of the selected design: _____

Write your Group ID: _____

Annotate, mark-up, and write on this page as needed.



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Ranking Order:

(4pts)

1. Rollover E
2. Low-Speed Maneuverability 3pts. E-F G-H
3. Top-Speed 2pts. Best
4. Towing Capacity 1pt. Worst

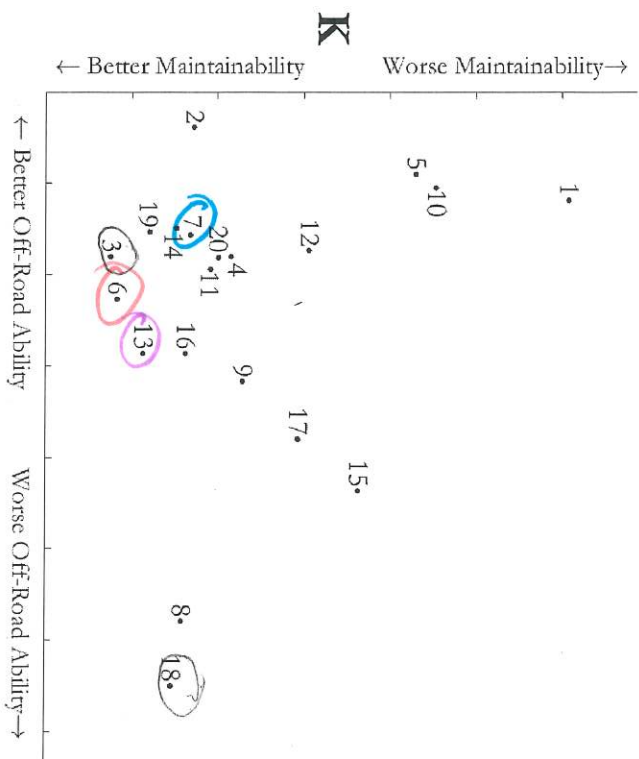
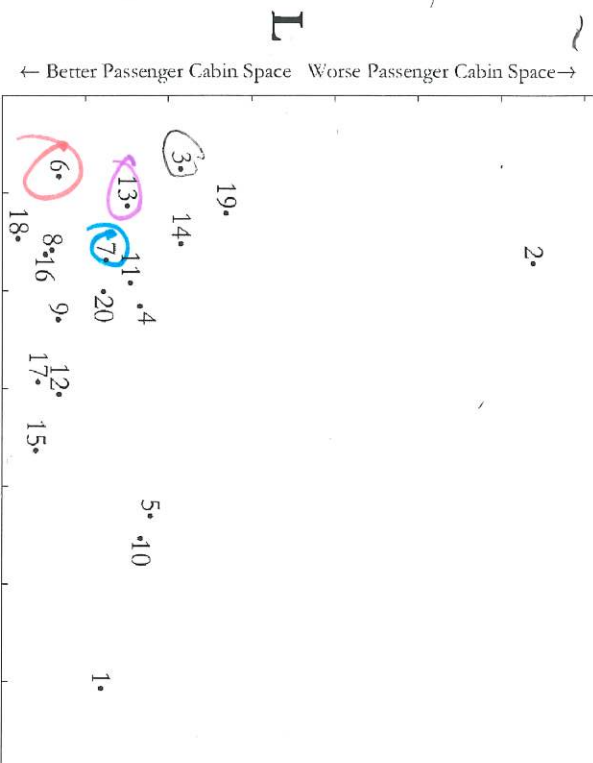
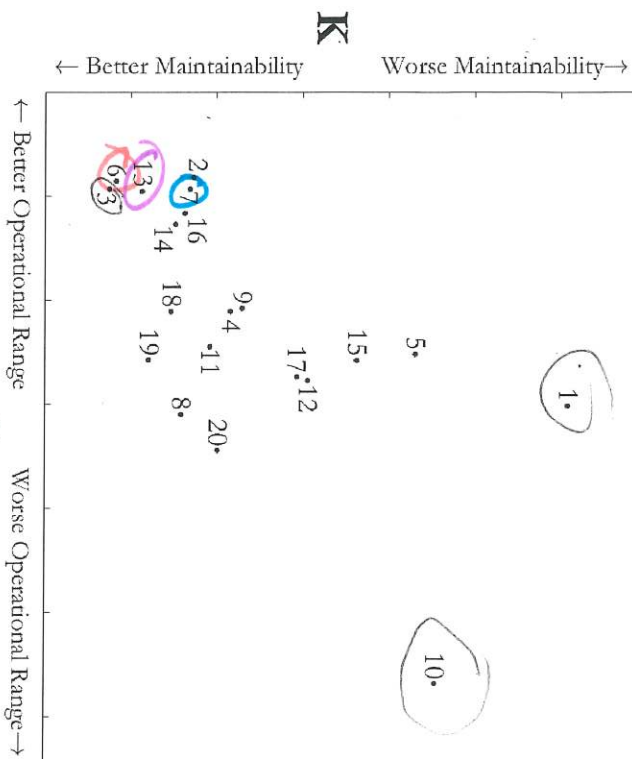
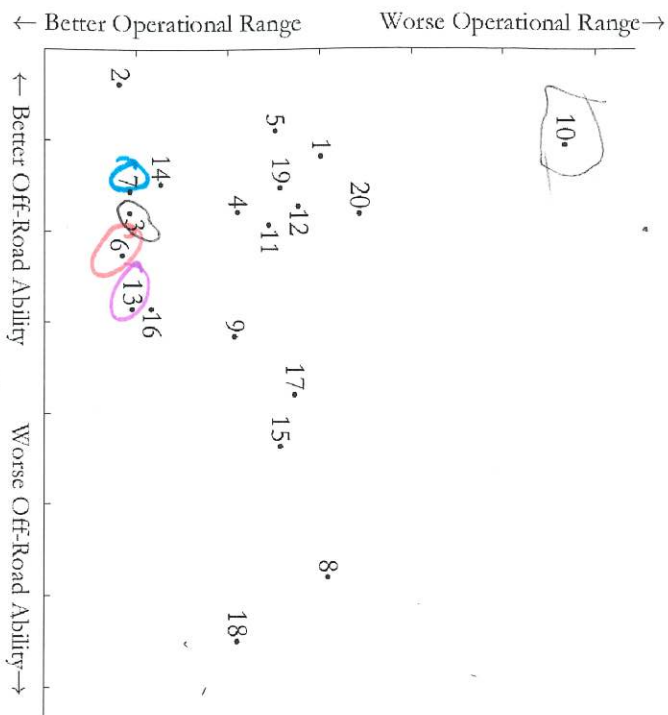
"Best" 20 12 3/11 a
 "Worst" 8 6 18 10

Speed	Weight
Rollover	3 11 5 20 2
Maneuver	3 11 20 14 2
Top Speed	8 9 15 4 6
Towing	9 12 1 16 8

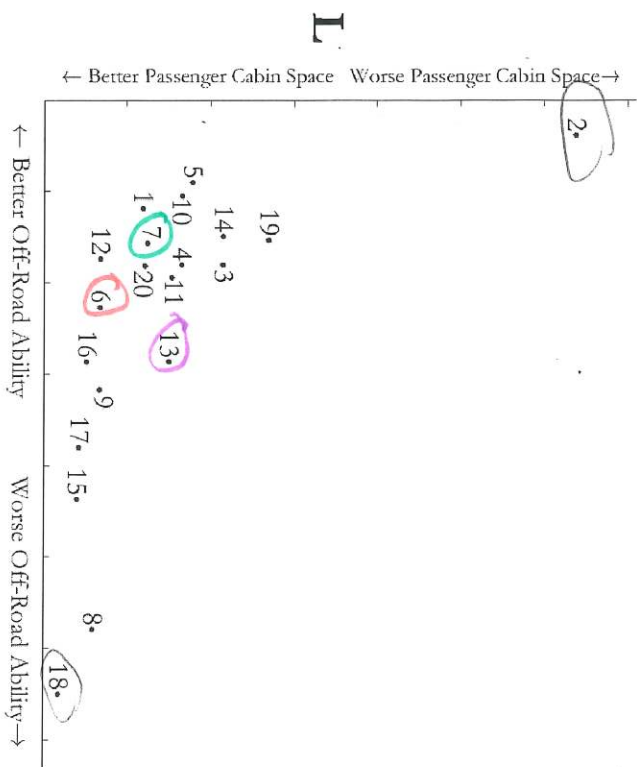
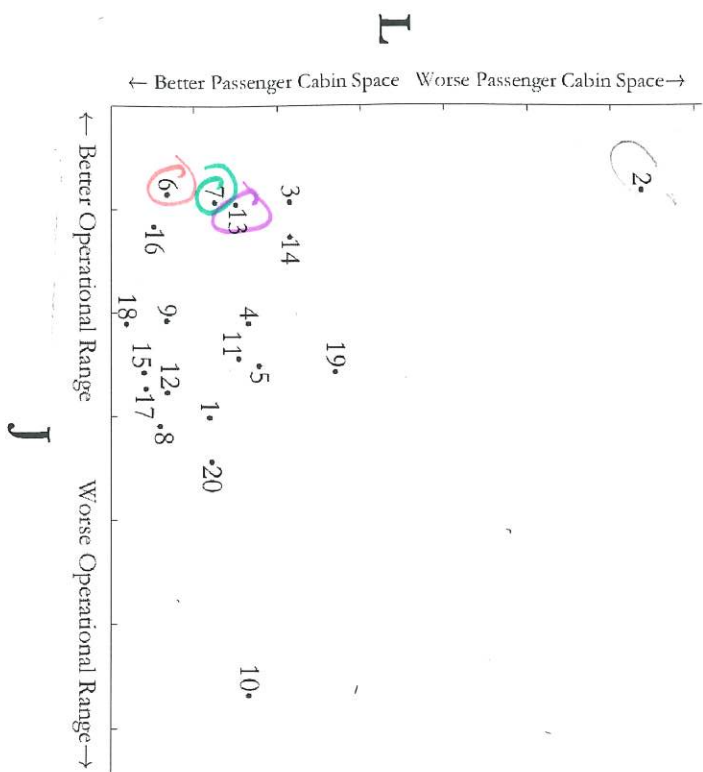
Enter the number ID of the selected design: _____

Write your Group ID: _____

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